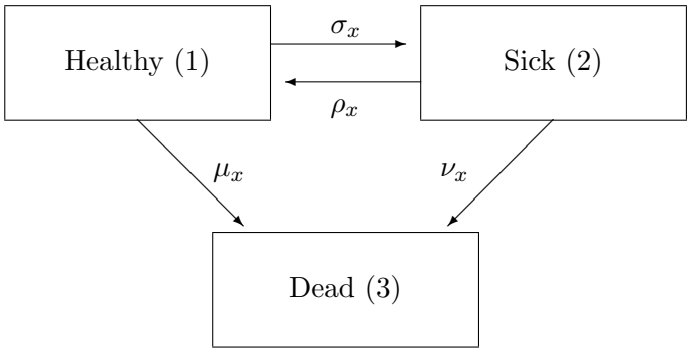
Stochastic Processes 2021/22 Semester 1

Tutorial 10

1. Consider the Markov model for sickness illustrated in the following diagram:



   (a) Differentiate with respect to u the formula of the lectures

$$p_{11}(x,u) = p_{\overline{11}}(x,u) + \int_x^u p_{12}(x,y)\,\rho_y\,p_{\overline{11}}(y,u)\,dy$$

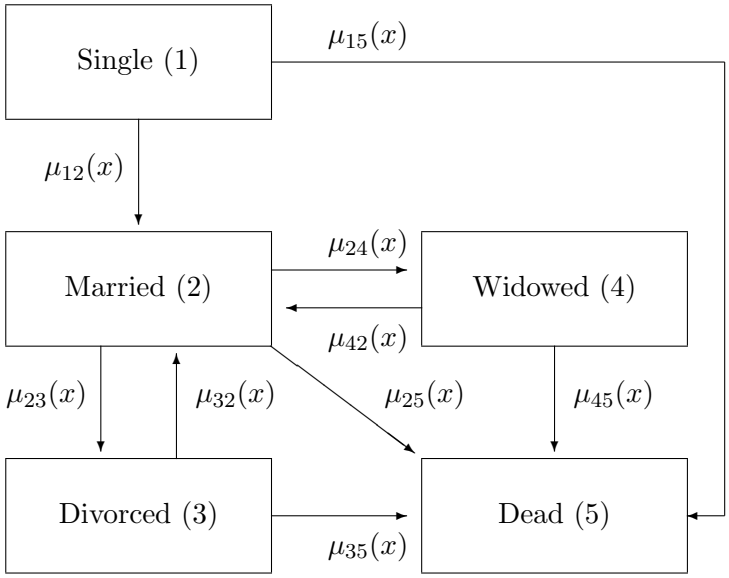
   and show that the derivative satisfies Kolmogorov's forward differential equation.

   (b) Explain why the following two formulae are correct:

$$\begin{aligned} p_{12}(x,u) &= \int_x^u p_{11}(x,y)\,\sigma_y\,p_{\overline{22}}(y,u)\,dy \\ p_{11}(x,u) &= p_{\overline{11}}(x,u) + \int_x^u p_{\overline{11}}(x,y)\,\sigma_y\,p_{21}(y,u)\,dy \end{aligned}$$

   (c) Differentiate with respect to u each of the two formulae in part (b) and show that they satisfy Kolmogorov's forward differential equations. You may use without proof Kolmogorov's forward differential equation for $p_{21}(y,u)$.

2. Consider the Markov model for marriage illustrated in the following diagram:



(a) i. Explain why, for $0 < t, h$,

$$\begin{aligned} p_{12}(x, x+t+h) = \; & p_{12}(x, x+t)\Big(1 - h(\mu_{24}(x+t) + \mu_{23}(x+t) + \mu_{25}(x+t))\Big) \\ & + p_{11}(x, x+t)h\mu_{12}(x+t) + p_{14}(x, x+t)h\mu_{42}(x+t) \\ & + p_{13}(x, x+t)h\mu_{32}(x+t) + o(h). \end{aligned}$$

ii. How can the formula in part (i), together with corresponding formulae for $p_{13}(x, x+t+h)$ and $p_{14}(x, x+t+h)$, be used to evaluate $p_{12}(x, x+t)$ numerically?

iii. Show that

$$\begin{aligned} \frac{\partial}{\partial t} p_{12}(x, x+t) = & -p_{12}(x, x+t)\Big(\mu_{23}(x+t) + \mu_{24}(x+t) + \mu_{25}(x+t)\Big) \\ & + p_{11}(x, x+t)\mu_{12}(x+t) + p_{14}(x, x+t)\mu_{42}(x+t) \\ & + p_{13}(x, x+t)\mu_{32}(x+t). \end{aligned}$$

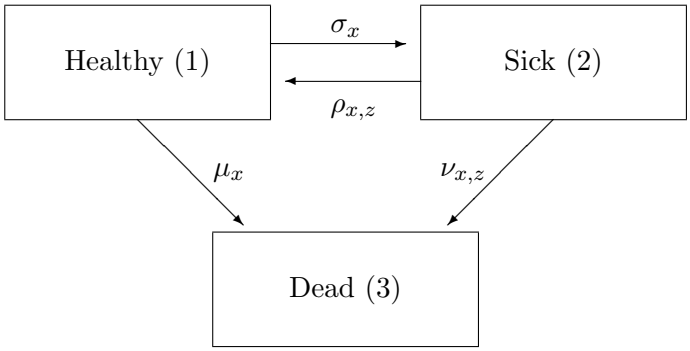
iv. Show that

$$\begin{aligned} p_{12}(x, x+t) = \int_0^t \Big(& p_{11}(x, x+u)\mu_{12}(x+u) + p_{14}(x, x+u)\mu_{42}(x+u) \\ & + p_{13}(x, x+u)\mu_{32}(x+u)\Big)\, p_{\overline{22}}(x+u, x+t)\, du. \end{aligned}$$

(b) Explain why

$$p_{12}(x, x+t) = \int_0^t p_{11}(x, x+u)\mu_{12}(x+u)\, p_{22}(x+u, x+t)\, du.$$

3. Consider the semi-Markov model for sickness illustrated in the following diagram, where x denotes the individual's current age and z denotes the duration of the current sickness.



Let

$$
\begin{aligned}
p_{11}(x, x+t) &= \mathbf{P}(\text{healthy at age } x+t \mid \text{healthy at age } x) \\
p_{12}(x, x+t; w) &= \mathbf{P}(\text{sick at age } x+t \text{ with duration} \leq w \mid \text{healthy at age } x) \\
p_{\overline{22}}(x, x+t) &= \mathbf{P}(\text{sick at age } x+t \text{ with duration } t \mid \text{became sick at age } x).
\end{aligned}
$$

(a) Explain why

$$
\frac{\partial}{\partial w} p_{12}(x, x+t; w) = \begin{cases} p_{11}(x, x+t-w)\sigma_{x+t-w}\, p_{\overline{22}}(x+t-w, x+t) & \text{for } 0 \leq w < t \\ 0 & \text{for } w > t. \end{cases}
$$

(b) Explain why

$$
\begin{aligned}
\frac{\partial}{\partial t} p_{11}(x, x+t) = &-p_{11}(x, x+t)(\mu_{x+t} + \sigma_{x+t}) \\
&+ \int_0^t p_{11}(x, x+u)\sigma_{x+u}\, p_{\overline{22}}(x+u, x+t)\rho_{x+t,t-u}\, du.
\end{aligned}
$$